	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 3

Lab Title Functions in MatLab

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Objective:

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____



Department of Electrical and Computer Engineering

Signal and Systems Lab

Lab Report: 3

Title: Introduction to

Loops and Selection

Control Structures

Name: Ali Abbas Khan

Roll Number: 230661

Class: BCE-5B

Submitted To: Engr. Maryam
Date: October 4, 2025

Lab Tasks

1. Create a function that transform years in days. Call this function in script file, input the number of years, and get the number of days as output.

Function Definition:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L3\ytod.m

ytod.m | Task1.m

```
1   function y=ytod(d)
2 -   y=d*365;
3 -   end
```

Function Call:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L3\Task1.m

ytod.m | Task1.m

```
1 -   clc; clear;
2 -   years=input('Enter the Years');
3 -   days=ytod(years);
4 -   fprintf("Days: %d\n",days);
```

Output:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L3\Task1.m

Command Window

```
Enter the Years8
Days: 2920
fx >>
```

2. Write a function fahrenheitToCelsius(F) that converts temperature from Fahrenheit to Celsius. Use Script file to Call the function Use the formula: C = (F - 32) * 5/9.

Function Definition:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L3\farenhietTocelcius.m

farenhietTocelcius.m | Task2.m

```
1   function C=farenhietTocelcius(F)
2 -   C=(F-32)*5/9;
3 -   end
```

Function Call:

```
Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L3\Task2.m
farenhietTocelcius.m | Task2.m
1 -  clc;
2 -  E=input('Enter Temperature in Farenhiet: ');
3 -  V=farenhietTocelcius(E);
4 -  fprintf("Temperature in Farenhiet: %f\n",V);
```

Output:

```
Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L3\Task2.m
Command Window
Enter Temperature in Farenhiet: 100.5
Temperature in Farenhiet: 38.055556
fx >>
```

3. Create a function in MATLAB which displays the following output:
a) The entered number is a positive/negative number.
b) The entered number is even/odd number.
Call this function in script file and use input command to take input number from user.
Function Call & Definitions:

```
Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L3\Mainfunction.m
Mainfunction.m
1   function Mainfunction()
2 -   Num=input('Enter a Number: ');
3 -   Even_Odd(Num);
4 -   Pos_Neg(Num);
5 -   end
6
7   function Even_Odd(Num)
8 -   d=mod(Num,2);
9 -   if d==0
10 -      fprintf("Even Number\n");
11 -  else
12 -      fprintf("Odd Number\n");
13 -  end
14 -  end
15
16  function Pos_Neg(num)
17 -  if num<0
18 -      fprintf("Negative Number\n");
19 -  elseif num>0
20 -      fprintf("Positive Number\n");
21 -  else
22 -      fprintf("Zero\n");
23 -  end
24 -  end
```

Output:



Conclusion:

Functions in MATLAB are reusable blocks of code designed to perform specific tasks, making programs more organized, efficient, and easier to debug. They allow users to encapsulate logic—such as mathematical operations, condition checks, or data transformations—into named routines that can be called from script files or other functions. Each function typically accepts input arguments, processes them, and returns output values. For example, in this lab, we created functions to convert years into days, transform Fahrenheit to Celsius, and classify numbers as positive/negative and even/odd. By using functions, we reduce repetition, improve clarity, and enable modular programming, which is essential for building scalable and maintainable code.